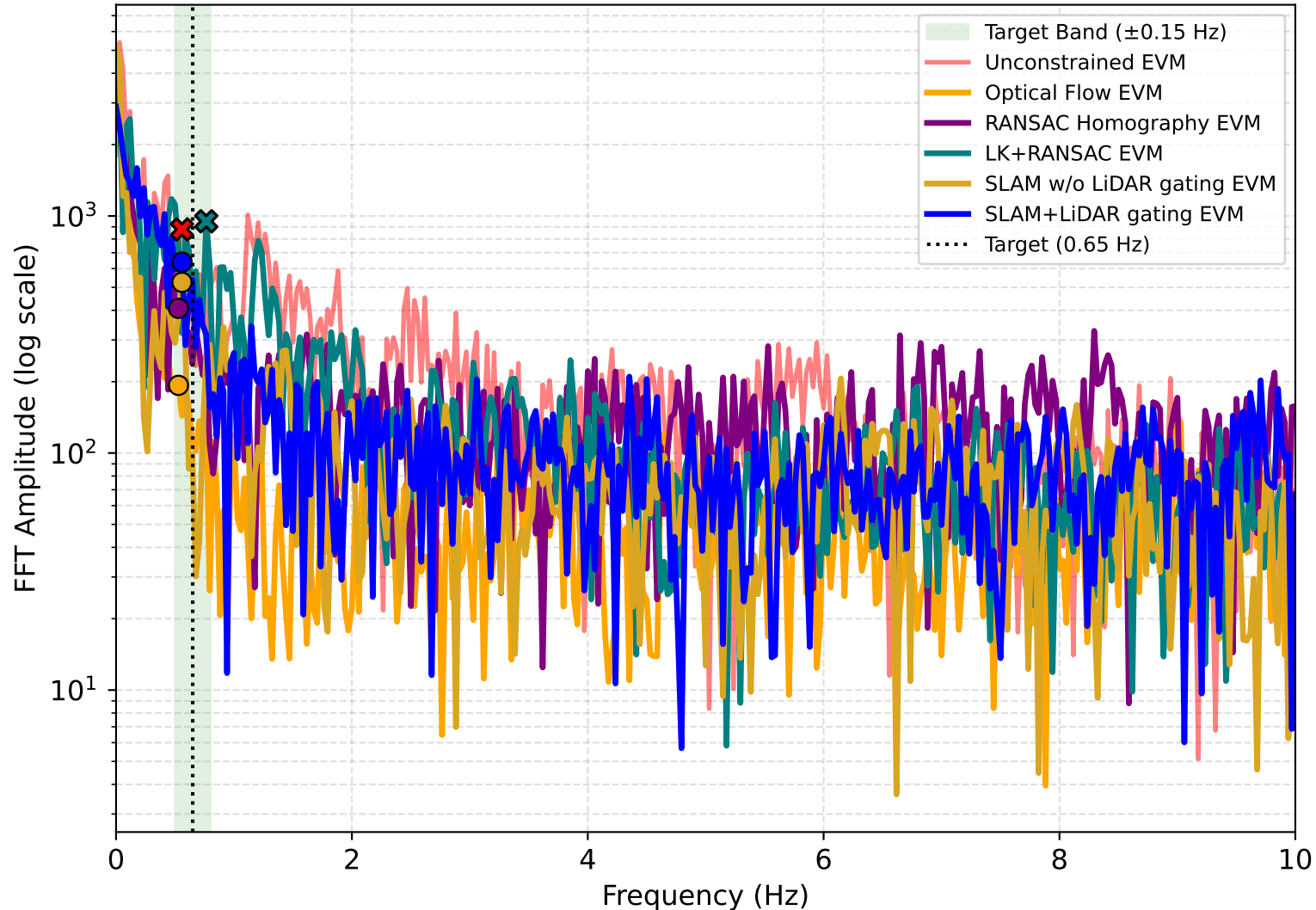


Vibration Spectrum Extraction Comparison (Log Scale, Target-Frequency Peaks Annotated)



Peak @ target band, by SNR:

LK+RANSAC
950 amp, 20.6 dB
△ UNRELIABLE (33% frames off-target, see Fig 2)
SLAM+LiDAR gating
639 amp, 17.2 dB
0% frames off-target (validated)
SLAM w/o LiDAR gating
525 amp, 17.2 dB
0% frames off-target (validated)
Unconstrained
878 amp, 16.6 dB
△ UNRELIABLE (60% frames off-target, see Fig 2)
Optical Flow
193 amp, 12.2 dB
7% frames off-target (validated)
RANSAC Homography
408 amp, 9.7 dB
7% frames off-target (validated)

Flag threshold: off-target ($>125\text{px}$, $\frac{1}{2} \text{ROI_BOX_SIZE}$) on $>15\%$ of frames
marker: ● validated × unreliable